

Derivación Matemática del modelo

1. Umbral bajo producción 2010 2. Umbral bajo ITA consorcional

$$V^S = w + s\mu - \frac{ps^2}{2\pi} - b \quad V^C = V^S + \varphi^S - r_c$$

$$V^S \geq 0 \quad V^C = w + s\mu - \frac{ps^2}{2\pi} - b + \varphi^S - r_c$$

$$w + s\mu - \frac{ps^2}{2\pi} - b \geq 0 \quad V^C \geq 0$$

$$w + s\mu - \frac{ps^2}{2\pi} \geq b$$

$$w - \frac{ps^2}{2\pi} - s\mu + \frac{ps^2}{2\pi} \geq b - s\mu$$

$$w \geq b - s\mu + \frac{ps^2}{2\pi}$$

$$V^S \geq 0 \Leftrightarrow w \geq TS$$

$$w \geq TS - (C_S - r_c)$$

$$\frac{TC}{V^C} = \frac{TS - (C_S - r_c)}{V^C} \Leftrightarrow w \geq \frac{TC}{V^C}$$

3. Umbral efectivo antes de Claude Code

$$V^1 \geq 0 \Leftrightarrow \max\{V^S, V^C\} \geq 0$$

$$V^1 = \max\{V^S, V^C\}$$

$$\rightarrow (V^S \geq 0) \Leftrightarrow (V^C \geq 0) \Leftrightarrow (w \geq TS)$$

$$(w \geq T^c)$$

$$(w \geq TS) \Leftrightarrow (w \geq T^c) \Leftrightarrow w \geq \min\{TS, T^c\}$$

$$T^1 = \min\{TS, TS - (C_S - r_c)\}$$

$$T^1 = TS - \max\{0, C_S - r_c\}$$

$$\text{Si } \varphi^S - r_c \leq 0 \rightarrow \max\{0, \varphi^S - r_c\} = 0$$

$$\rightarrow T^1 = TS$$

$$V^D = w + (1-\lambda)S\mu + \lambda\alpha\pi(A) - K(C, S, A) - K(C, \alpha, S) - r_D - b$$

$$- \frac{p}{2} \left[(1-\lambda)^2 \frac{S^2}{\pi} + \sigma_D^2(C, S, A) \right] \geq 0$$

$$w + (1-\lambda)S\mu + \lambda\alpha\pi(A) - K(C, S, A) - r_D - b - \frac{p}{2} \left[(1-\lambda)^2 \frac{S^2}{\pi} + \sigma_D^2(C, S, A) \right] \geq 0$$

$$w + (1-\lambda)S\mu + \lambda\alpha\pi(A) - K(C, S, A) - r_D - \frac{p}{2} \left[(1-\lambda)^2 \frac{S^2}{\pi} + \sigma_D^2(C, S, A) \right] \geq b$$

$$w + (1-\lambda)S\mu + \lambda\alpha\pi(A) - K(C, S, A) - \frac{p}{2} \left[(1-\lambda)^2 \frac{S^2}{\pi} + \sigma_D^2(C, S, A) \right] \geq b + r_D$$

$$w \geq b + r_D + K(C, \alpha, S) - (1-\lambda)S\mu - \lambda\alpha\pi(A) + \frac{p}{2} \left[(1-\lambda)^2 \frac{S^2}{\pi} + \sigma_D^2(C, S, A) \right]$$

$$T^D = b - (1-\lambda)S\mu - \lambda\alpha\pi(A) + K(C, S, A) + r_D + \frac{p}{2} \left[(1-\lambda)^2 \frac{S^2}{\pi} + \sigma_D^2(C, S, A) \right]$$

$$V^D \geq 0 \Leftrightarrow w \geq T^D$$

5. Umbral efectivo después de clave code

$$M^2 = h(s, c, D)$$

$$V^2 = \max(V^S, V^P, V^D)$$

$$V^2 \geq 0 \Leftrightarrow \max(V^S, V^P, V^D) \geq 0$$

$$(V^S \geq 0) \vee (V^P \geq 0) \vee (V^D \geq 0)$$

$$\rightarrow (w \geq T^S) \vee (w \geq T^C) \vee (w \geq T^D) \Leftrightarrow \min\{T^S, T^C, T^D\}$$

$$T^R = \min\{T^S, T^C, T^D\}$$

$$T^P = \min\{T^S, T^D\}$$

$$T^1 = \min\{T^S, T^C\}$$

$$T^2 \leq T^1$$

6. Derivación de $B = T^1 - T^D$

Para lenguaje desconocido: $T^1 = T^S$

$$B = T^1 - T^D$$

$$B = T^S - T^D$$

$$T^S = b - s\mu + \frac{ps^2}{2\pi}$$

$$T^D = b - (1-\lambda)s\mu - \lambda a\tau(A) + \kappa(a, s) + r_D$$

$$+ \frac{p}{2} \left[\frac{(1-\lambda)^2 s^2}{\pi} + \sigma_0^2(a, s, A) \right]$$

$$B = \left(b - s\mu + \frac{ps^2}{2\pi} \right) - \left[b - (1-\lambda)s\mu - \lambda a\tau(A) + \kappa(a, s) + r_D + \frac{p}{2} \left[\frac{(1-\lambda)^2 s^2}{\pi} + \sigma_0^2(a, s, A) \right] \right]$$

$$\theta = \lambda [a\tau(A) - s\mu] - \kappa(a, s) - r_D + \frac{p}{2} \left[(2\lambda - 1)^2 \frac{s^2}{\pi} - \sigma_0^2(a, s, A) \right]$$

7. De $\theta > 0$ a la banda de activación

$$\theta = T^S - T^D > 0 \Rightarrow T^S > T^D \quad ; \quad T^1 = T^S$$

$$T^D < T^1 \rightarrow T^2 = \min\{T^1, T^D\}$$

$$T^D < T^1 \rightarrow T^P = T^D$$

$$T^1 = 1[w \geq T^2]$$

$$T^2 = 1[w \geq T^2]$$

Como $T^1 = T^S$ y $T^2 = T^D$

$$T^1 = 1[w \geq T^S]$$

$$T^2 = 1[w \geq T^D]$$

$$T^2 - T^1 = 1[w \geq T^D] - 1[w \geq T^S]$$

Si $w < T^D \rightarrow w < T^S$ $T^2 - T^1 = 0 - 0 = 0$

Si $w < T^D \leq T^S \rightarrow T^2 - T^1 = 1 - 0 = 1$

Si $w \geq T^S \rightarrow (T^D > T^D) \rightarrow w \geq T^D$ $T^2 - T^1 = 1 - 1 = 0$

o.o $T^P - T^1 = 1[T^D \leq w < T^S]$

8. Cambio total en el número de lenguajes

$$N^2 = \sum_R Z_R^S$$

Para lenguajes con $B_R > 0$ $Z_R^2 - Z_R^1 = 1[T_R^D \leq w_R < T_R^S]$

$$\rightarrow N^2 - N^1 = \sum_R 1[T_R^D \leq w_R < T_R^S]$$

9. Esperanza en cambio en lenguajes

$$E[N^2 - N^1] = E\left[\sum_k (Z_k^2 - Z_k^1)\right]$$

$$E[N^2 - N^1] = \sum_k E[Z_k^2 - Z_k^1] \quad Z_k^g = 1[w_k \geq T_k^g]$$

$$E[Z_k^g] = E[1[w_k \geq T_k^g]]$$

$$E[1[w_k \geq T_k^g]] = \Pr(w_k \geq T_k^g)$$

$$E[Z_k^g] = \Pr(w_k \geq T_k^g)$$

$$\text{Sea } F_k(x) = \Pr(w_k \leq x) \rightarrow \Pr(w_k \geq T_k^g) = 1 - F_k(T_k^g)$$

$$E[Z_k^2 - Z_k^1] = [1 - F_k(T_k^2)] - [1 - F_k(T_k^1)]$$

$$E[Z_k^2 - Z_k^1] = F_k(T_k^1) - F_k(T_k^2)$$

$$E[N^2 - N^1] = \sum_k (F_k(T_k^1) - F_k(T_k^2))$$

Como $T_k^2 \leq T_k^1$ y $F_k(\cdot)$ es no decreciente

$$\Rightarrow F_k(T_k^2) \leq F_k(T_k^1)$$

$$F_k(T_k^1) - F_k(T_k^2) \geq 0$$

$$E[N^2 - N^1] \geq 0$$

10. Forma específica para la banda de activación

Para lenguaje desconocido: $T_k^g = T_k^s$

$$\text{Si } B_k > 0 \rightarrow T_k^c = T_k^0$$

$$\text{Wego: } E[Z_k^2 - Z_k^1] = F_k(T_k^s) - F_k(T_k^0)$$

Sumando sobre lenguajes desconocidos $\in \mathcal{L}_k$:

$$E[N_k^2 - N_k^1] = \sum_{k \in \mathcal{L}_k} [F_k(T_k^s) - F_k(T_k^0)] \geq 0$$

11. Finalmente

$$v^0 \geq 0 \leftrightarrow w \geq T^0$$

$$b = T^s - T^0$$

$$b > 0 \rightarrow T^0 < T^s$$

$$\Rightarrow Z^2 - Z^1 = 1[T^0 \leq w < T^s]$$

$$N^2 - N^1 = \sum_k (Z_k^2 - Z_k^1)$$

$$E[N^2 - N^1] = \sum_k (F_k(T_k^1) - F_k(T_k^2)) \geq 0$$

$$\text{Claude code} \rightarrow T^0 < T^s \rightarrow Z^2 - Z^1 = 1[T^0 \leq w < T^s] \rightarrow E[N^2 - N^1] \geq 0$$